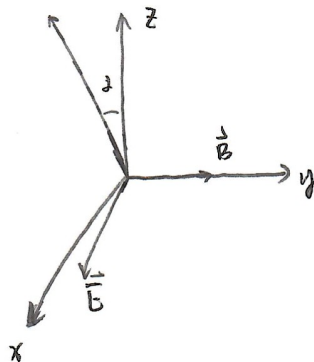


10.19 (a) For polarization in the plane of incidence, as shown in the right, we have



$$\vec{E}_0 = -E_0 \sin \alpha \vec{e}_3, \quad \vec{B}_0 = \frac{E_0}{c} \vec{e}_y.$$

From Prob. 10.18, the angular distribution of the transmitted power is

$$\frac{dP_{||}}{d\Omega} = \frac{1}{18\pi^2 Z_0} k^4 a^6 \left[4E_0^2 (\cos^2 \theta + \sin^2 \theta \cos^2 \phi) + E_0^2 \sin^2 \alpha \sin^2 \theta + 4E_0^2 \sin \alpha \sin \theta \cos 2\theta \cos \phi \right]$$

and total transmitted power is

$$P_{||} = \frac{2}{27\pi Z_0} k^4 a^6 E_0^2 (4 + \sin^2 \alpha).$$

For polarization perpendicular to the incident plane, we have

$$\vec{E}_0 = 0, \quad \vec{B}_0 = -\frac{E_0}{c} \cos \alpha \vec{e}_1.$$

Repeat the same calculation as in Prob. 10.18, we have

$$\frac{dP_{\perp}}{d\Omega} = \frac{2}{9\pi^2 Z_0} k^4 a^6 E_0^2 \cos^2 \alpha (\cos^2 \theta + \sin^2 \theta \sin^2 \phi).$$

and

$$P_{\perp} = \frac{8}{27\pi Z_0} k^4 a^6 E_0^2 \cos^2 \alpha.$$

(c). The total power normally incident on the aperture is

$$P_i = \frac{E_0^2}{2Z_0} \pi a^2 \cos \alpha$$

By definition,

$$T_{||} = \frac{P_{||}}{P_i} = \frac{4}{27\pi} (ka)^4 \frac{4 + \sin^2 \alpha}{\cos \alpha},$$

$$T_{\perp} = \frac{P_{\perp}}{P_i} = \frac{16}{27\pi} (ka)^4 \cos \alpha.$$

The difference in the prefactor is very likely due to the mix of SI and cgs. units.